Spatial omics, hands-on: one disease, three technologies

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ASI-FIMSA 2026 · two-hour Workshop · everything runs in Google Colab

Four Tutorials, two hours, one disease seen three ways

Notebook	What you do	Time
00 · Setup check	Confirms Colab can install the stack and reach the data. Run it at home, not in the room.	~10 min
01 · Read and visualise	Load and plot Visium spots, Xenium cells and the whole-transcriptome Atera run.	~35 min
02 · Niche analysis	Find the niches of a tumour microenvironment from which cells sit next to which.	~35 min
03 · Cell–cell interaction	LIANA+'s consensus ligand–receptor test on the same cells.	~30 min
04 · ViT for gene expression	Train a Vision Transformer that predicts expression from H&E alone.	~40 min

Every Tutorial is self-contained and one click from the README:
github.com/xiao233333/ASI-FIMSA-workshop-2026

One click opens it; Runtime → Run all does the rest

1
Click the badge

One per Tutorial in the README. Opens in your browser.

2
Runtime → Run all

Then wait. There is nothing else to click.

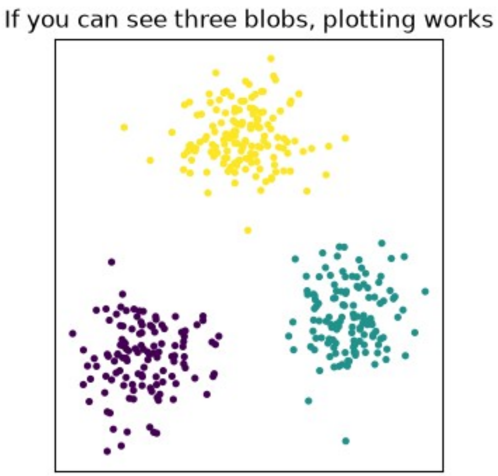
3
The install cell

About two minutes. It runs once per session.

4
Read downwards

Every cell is explained in the text above it.

- You need a Google account. Colab is free and needs no card.
- Run 00 · Setup check at home. It downloads about 300 MB, which conference wifi may not enjoy.
- Nothing is saved to your Google Drive unless you ask for it.



Setup check, Step 3b

If pip replaces a package, restart the session

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PLEASE RESTART THE SESSION, THEN RUN THIS NOTEBOOK AGAIN FROM THE TOP.
Use the menu:  Runtime → Restart session
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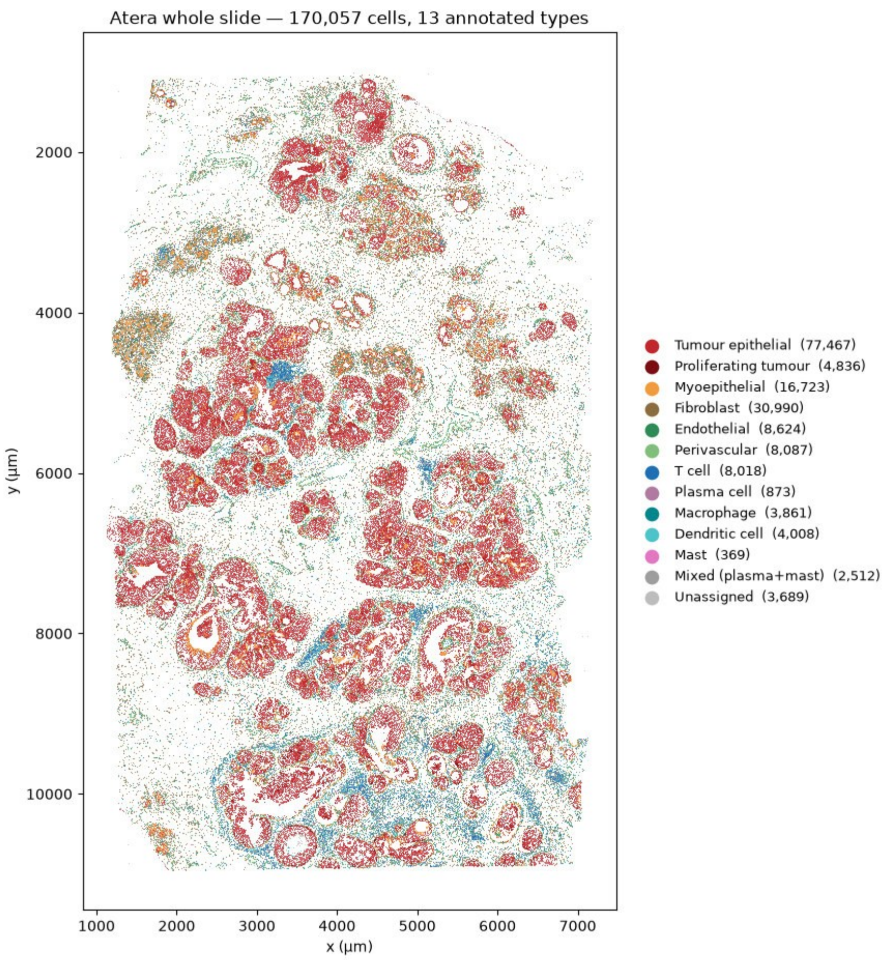
Installing the stack upgrades pandas, and Colab has already imported it. Every install cell watches for this and prints the banner itself.

- A GPU is optional. Every Tutorial runs on the ordinary free CPU runtime; Tutorial 4 is a few minutes quicker with one. If you want it: Runtime → Change runtime type → T4 GPU.
- Sessions are disposable. Close the tab, open the Tutorial again, start over. Nothing on your own machine can be broken by anything here.
- Colab wipes what you install when a session ends, so you run an install cell at the start of each Tutorial. That is expected, not a mistake.
- On the day, Tutorial 4 pulls a further 1.8 GB full-resolution H&E from 10x over Google's network, not the wifi in the room.

Tutorial 1: dissociation answers what is there, never where



- 170,057 cells, 13 annotated types, every one of them placed. Dissociate the same tissue and you keep the types and lose the positions.
- The myoepithelial rim around each duct is the histological signature of ductal carcinoma in situ. No UMAP can show it.
- The immune infiltration is heterogeneous across a single slide: dense in places, absent in others.

Tutorial 1 §3.3 · Atera whole-transcriptome Xenium preview · CC BY 4.0, 10x Genomics

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Two families of technology: capture the tissue, or image it in place

	Sequencing-based	Imaging-based
How it works	mRNA is captured on a barcoded surface, then sequenced	probes bind transcripts and are imaged in place, cycle by cycle
One measurement is	a fixed-size spot, and whatever cells fall in it	one cell, segmented from the image
Which genes	the whole transcriptome, chosen by nobody	whatever is on the panel — until Atera
Here	Visium	Xenium, Atera

- An imaging panel is designed before the experiment and fixes what is measurable.
- Atera is the first of the imaging family to drop that constraint.

Everything lands in one SpatialData object, in one coordinate system

.images
Pixels

H&E and morphology stains, usually stored at several resolutions.

.shapes
Geometry

Visium spots and cell boundaries, as circles or polygons.

.points
Molecules

Individual transcripts, one row each, with coordinates.

.tables
Counts

An AnnData of cells by genes, annotating one of the elements above.

- The elements do not share a pixel grid. Each carries a transformation into a named coordinate system, and that is what lines them up.
- Xenium ships its polygons in micrometres while its "global" system is in camera pixels. Mixing the two silently misplaces every cell.
- import spatialdata_plot looks unused. It registers the .pl accessor onto every object, which is how the figures get drawn.

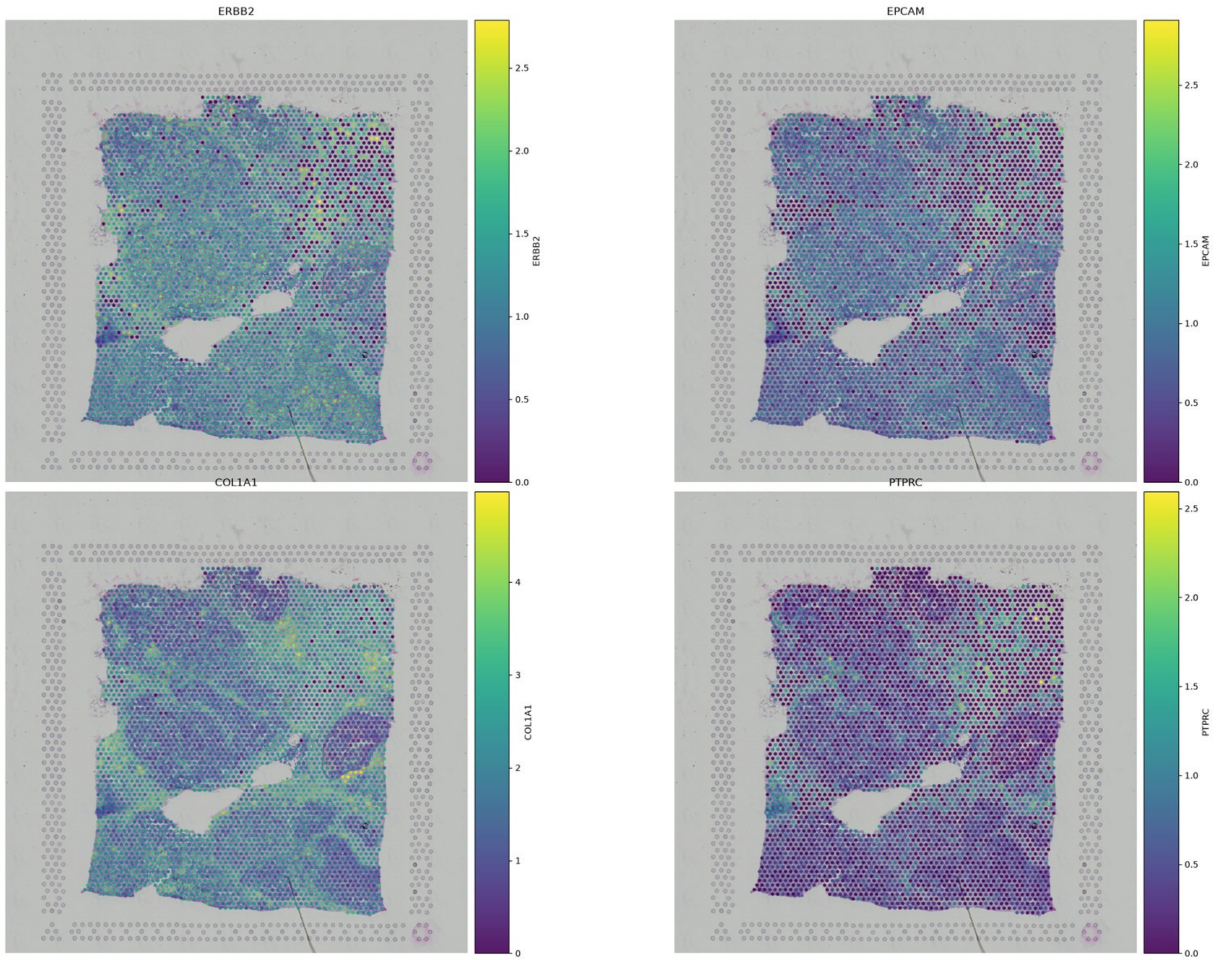
Every platform trades genes against cells: read the table as a diagonal

Platform	One measurement is	How many genes
Visium	a 55 µm spot: a small disc of tissue holding several cells	whole transcriptome, ~36,600
Xenium	one cell	a targeted panel, 313
Atera	one cell	whole transcriptome, ~18,000

Samples: V1_Breast_Cancer_Block_A_Section_1 · Xenium_FFPE_Human_Breast_Cancer_Rep1 · Atera WTA preview, Gen2 prototype

- Visium gives you every gene but not every cell. Xenium gives you every cell but only the genes someone chose in advance.
- Atera is the corner that used not to exist: single-cell resolution and the whole transcriptome, over 170,057 cells.
- Three different FFPE specimens of human breast carcinoma, not serial sections of one block. All three from 10x Genomics.

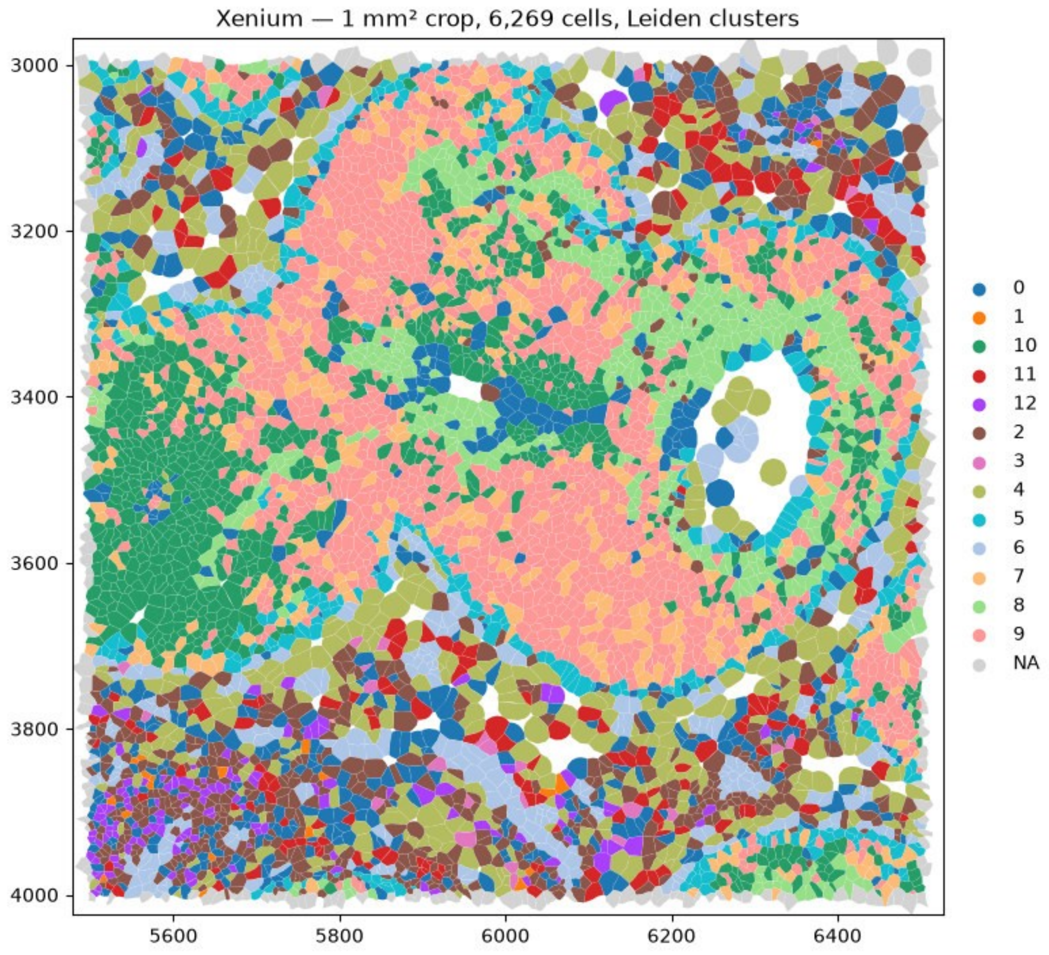
A 55 µm spot shows architecture nobody annotated, and hides everything inside it



- EPCAM and ERBB2 light up the same territory; COL1A1 fills the space between them. That is the tumour and stroma architecture of the section, visible before anyone annotated anything.
- Now look at PTPRC. Low and diffuse almost everywhere.
- Every spot is a mixture of cells, and you cannot co-localise within one.

Tutorial 1 §1.6 · Visium V1_Breast_Cancer_Block_A_Section_1, 3,798 spots under tissue · data: 10x Genomics

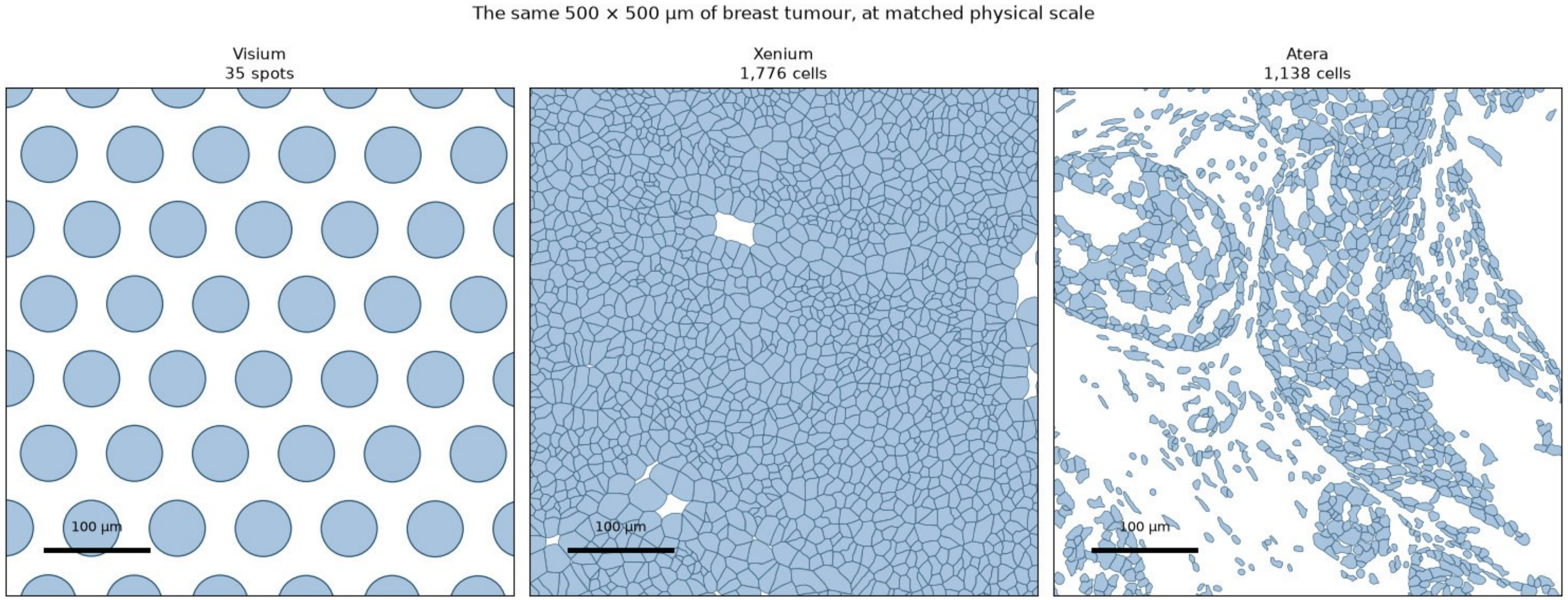
Xenium resolves single cells, so you can ask whether the T cells are inside the duct



- 6,269 cells in one square millimetre, clustered and drawn as their real segmentation boundaries.
- 313 genes, and 228 columns that are not genes: on-slide negative controls with no equivalent in a sequencing assay.
- Adjacency stops being a guess and becomes a question you can answer. That is Tutorial 2.

Tutorial 1 §2.8 · Xenium_FFPE_Human_Breast_Cancer_Rep1 · Janesick et al., Nat Commun 14, 8353 (2023)

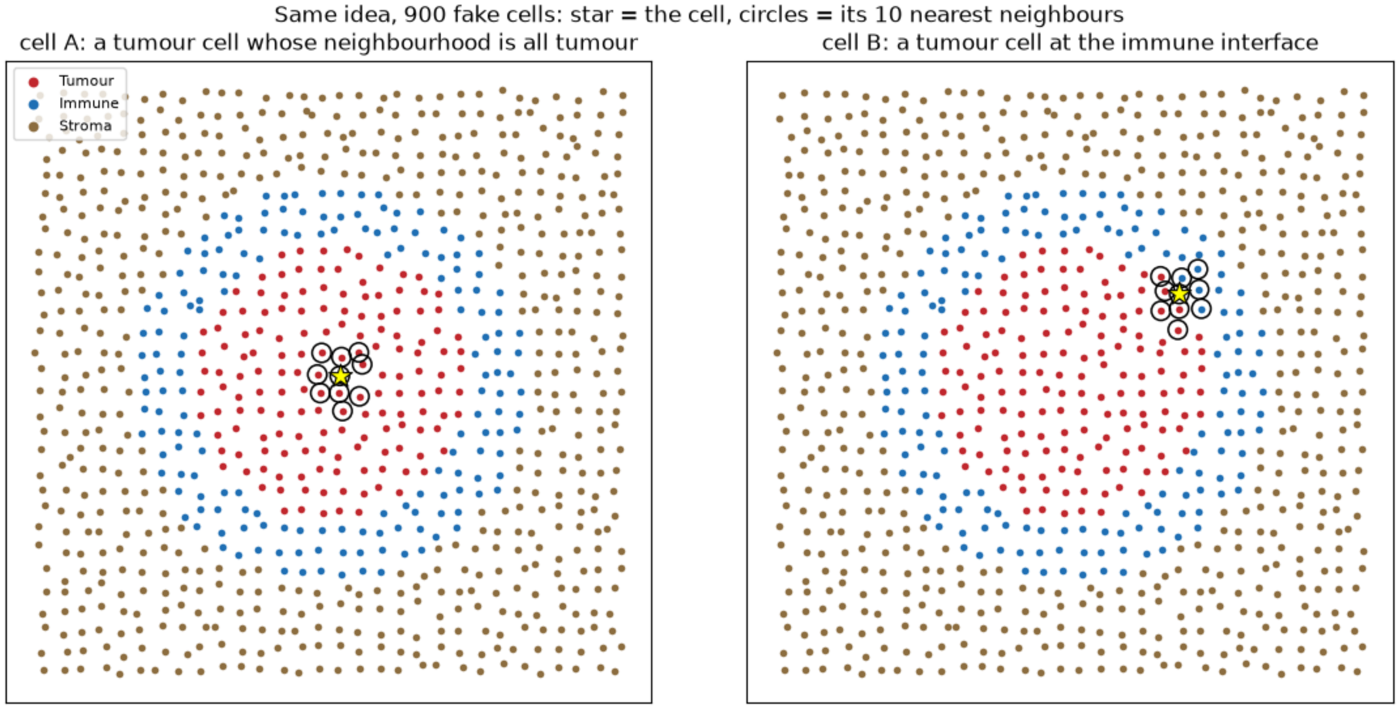
Atera gives both. But any one gene is sparse, and deconvolution is not resolution.



- The same 500 µm of tumour, three ways: 35 Visium spots, 1,776 Xenium cells, 1,138 Atera cells.
- Whole transcriptome at cell resolution is still sparse: EPCAM in ~47% of Atera cells, ERBB2 in ~4%.
- Xenium expands nuclei and tiles the plane; Atera leaves the extracellular space empty.

Tutorial 1 §4.2 · Atera whole-transcriptome Xenium preview, 18,028 genes, 170,057 cells · CC BY 4.0, 10x Genomics

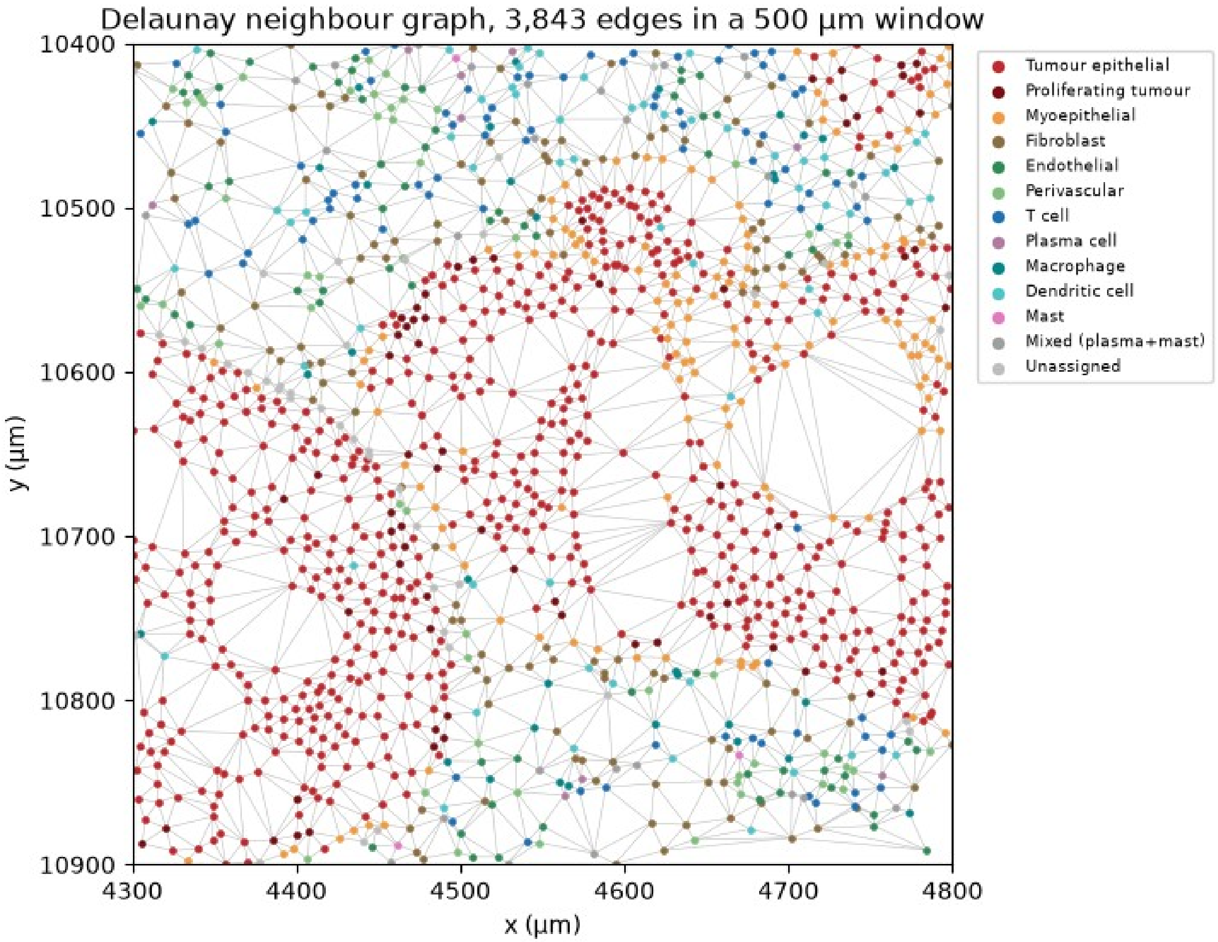
Tutorial 2: a cell-type list is a census, not an architecture



- Two tumour cells, same type, same expression. One is buried in tumour, the other sits at the immune interface.
- Nothing about the cells themselves separates them. Everything about their neighbourhoods does.
- Immune-excluded, inflamed and desert are claims about neighbourhoods. A table of cell-type counts cannot make them.

Tutorial 2 §2.1 · 900 synthetic cells, 10 nearest neighbours

Coordinates become a graph before they become a statistic



Tutorial 2 §3.1 · sq.gr.spatial_neighbors(delaunay=True)

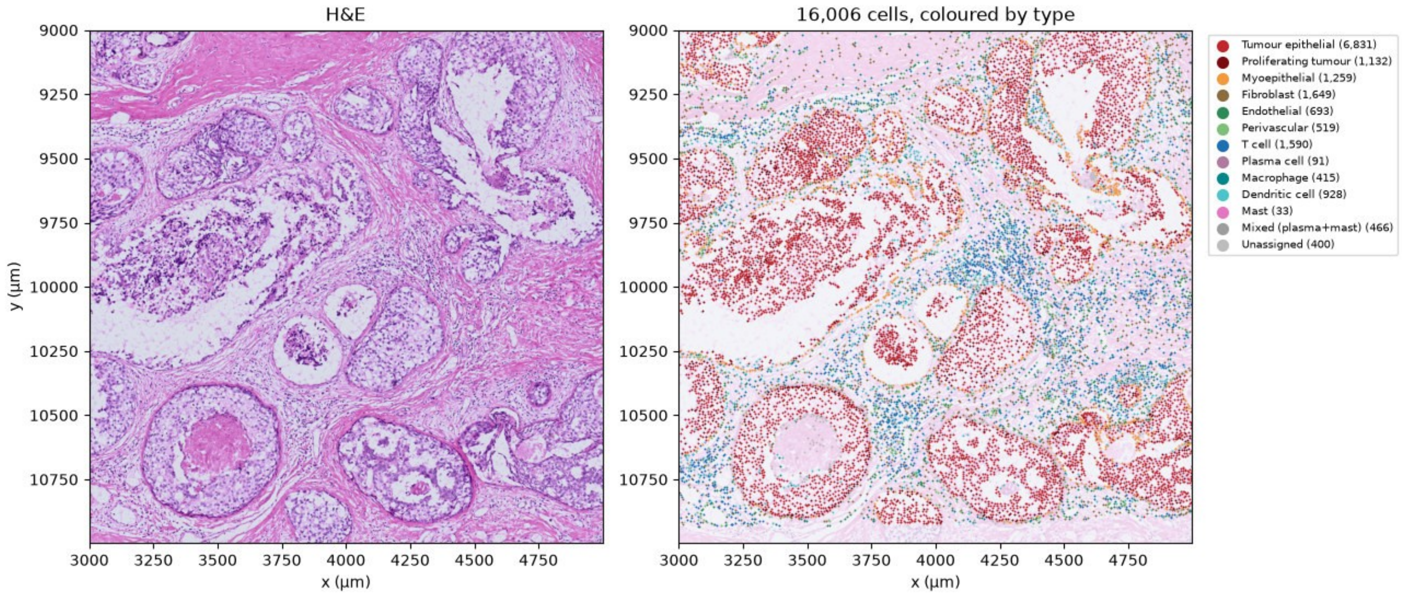
- Delaunay triangulation, a fixed radius, or k nearest neighbours. Each defines "neighbour" differently, and none is the right one.
- 3,843 edges in this 500 µm window. About 95% are shorter than 36 µm; the longest runs roughly 1,700 µm across a gap in the tissue.
- Look at the graph before you trust anything computed on it.

Three routes from a graph to a niche, answering three different questions

Route	What it computes	What comes back
squidpy nhood_enrichment	how often each pair of cell types is adjacent, against shuffled labels	one z-score per pair of types, for the whole Crop
kNN composition + KMeans	the mix of cell types around every cell, then clusters those profiles	a niche label for every single cell
sopa vectorize_niches	those labels turned into geometry	area, roundness, components, and distance in hops

- The first returns one number for the whole tissue. That is why the other two exist.
- All three take a scale parameter, and the answer changes with it. Report the number you used.

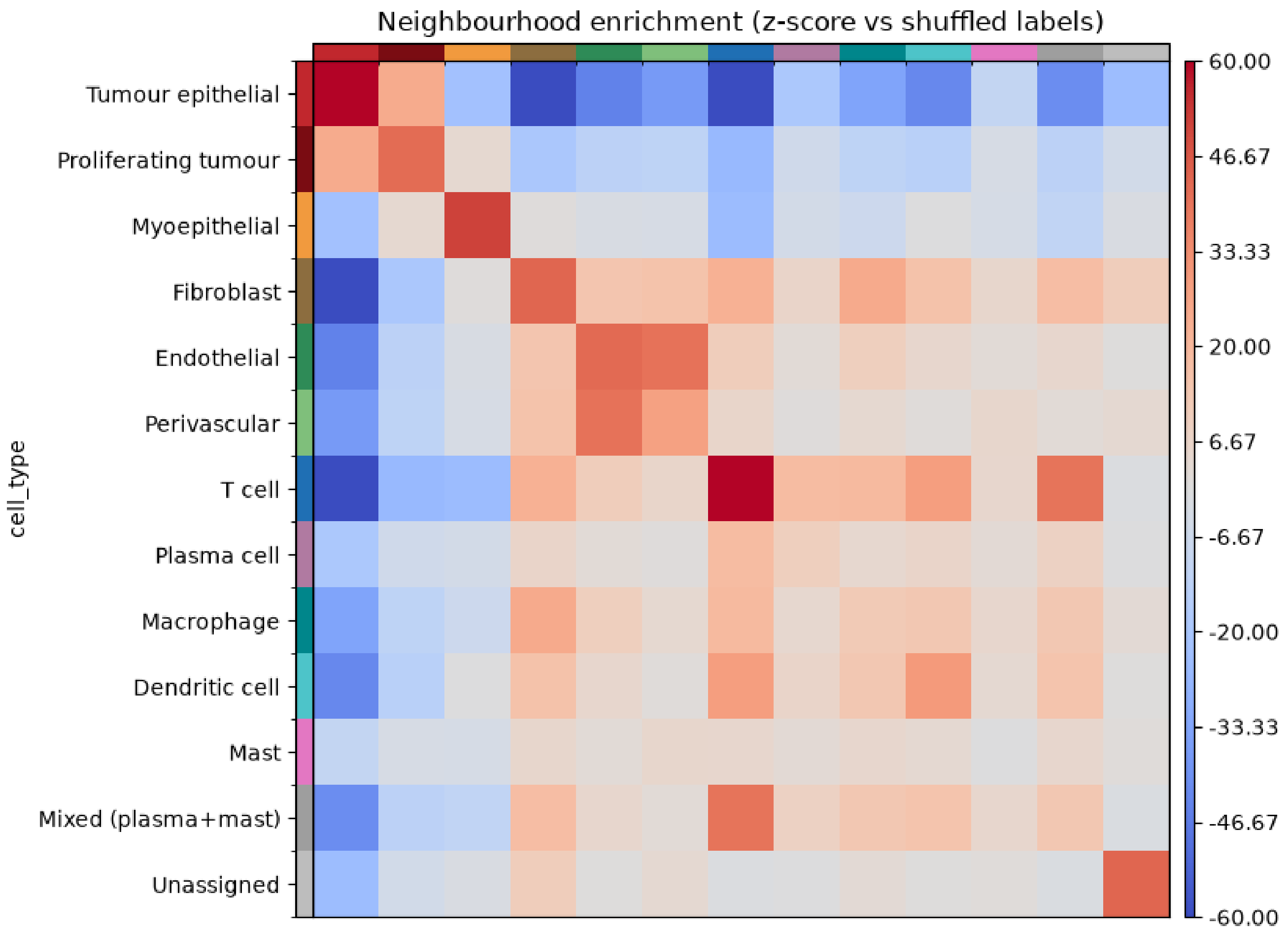
A cell-type list says who is in the room, not who is standing with whom



- The Crop: a 2,000 µm square, 16,006 cells, 12 named types. 43% is tumour epithelium.
- 400 cells stay Unassigned and are kept. Dropping them would quietly delete the cells the method understood least.
- Three questions follow: which types sit together, what niches exist, and how far apart they are.

Tutorial 2 §1.2 · Atera Crop, one 18 MB download · cell types from a human-reviewed annotation of 34 vendor clusters

Tumour epithelium avoids T cells at z ≈ −72: an immune-excluded tumour



- squidpy builds a Delaunay graph, then shuffles the labels 200 times to ask which pairs are adjacent more often than chance.
- Tumour self-enrichment is z ≈ +119. Against T cells −72, dendritic cells −43, macrophages −34.
- The lymphocytes are present, in decent numbers, and they are not getting in.
- Endothelial ↔ perivascular at +41 is the positive control: two types that must be adjacent, and are.

Tutorial 2 §3 · sq.gr.spatial_neighbors(delaunay=True) + sq.gr.nhood_enrichment, 200 permutations

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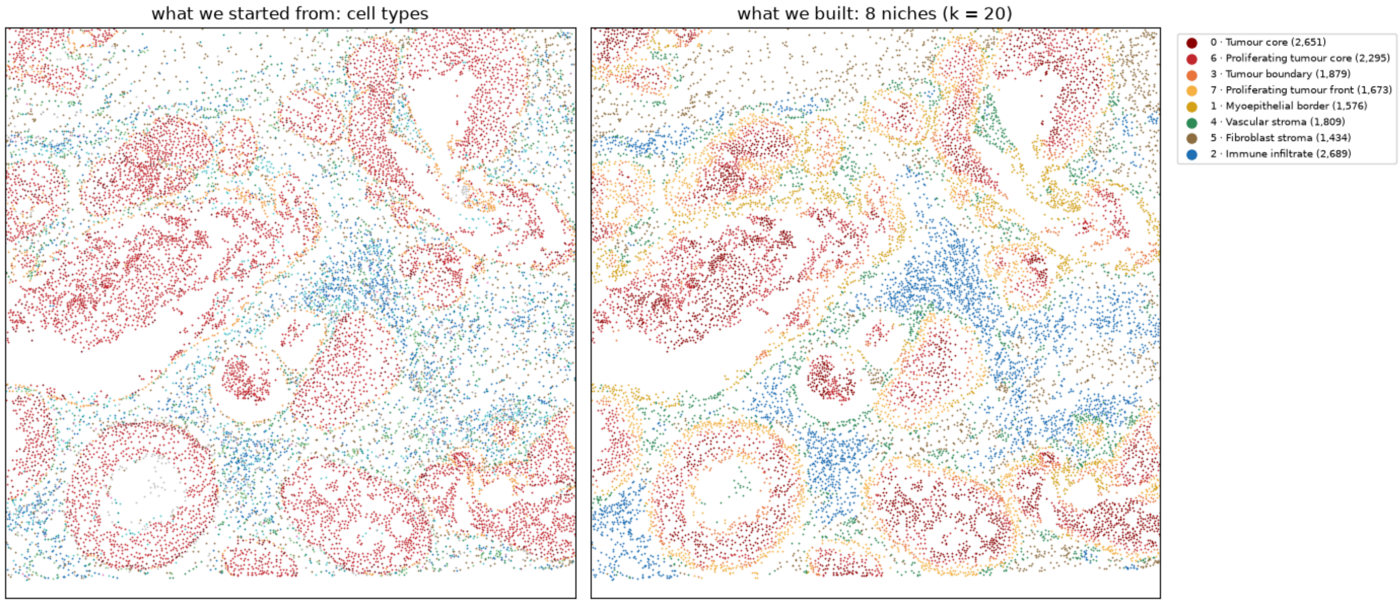
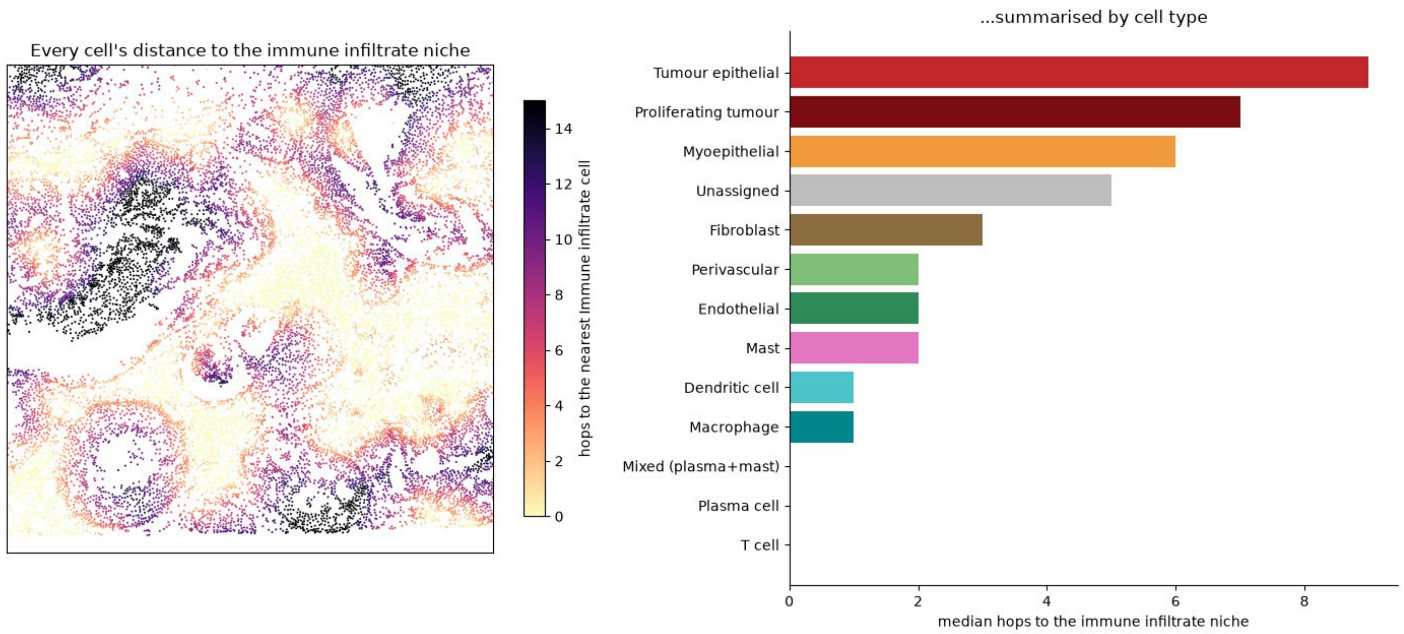
Niches built with no knowledge of position come out spatially coherent



- Neighbourhood composition from the 20 nearest cells (about 60 µm), then KMeans into 8 niches.
- The clustering never saw a coordinate, only the mix of types around each cell. The coherence was in the tissue.
- k is the spatial-scale knob. A niche is a property of a tissue at a stated scale.

Tutorial 2 §4.3 · kNN composition + MiniBatchKMeans · adapted from gml-teaching-2026, 3.2_neighborhood.ipynb

Tumour interiors sit 15 or more cell-to-cell steps from the nearest immune cell



- sopa turns each niche into a polygon you can measure: area, roundness, and distance in graph hops.
- Median hops to the immune infiltrate niche: tumour epithelial 9, proliferating tumour 7, T cells 0.
- Dendritic cells sit closer to the tumour niches than T cells do. This is "immune-excluded" as a number.

Tutorial 2 §5.4 · sopa.spatial.vectorize_niches / cells_to_groups / mean_distance

Tutorial 3: adjacency is not conversation

What is measured

Two mRNAs and a distance

Ligand mRNA in one population, receptor mRNA in another, and how far apart the cells sit.

What is claimed

A signalling axis

That these two populations are communicating through this pair, somewhere in this tissue.

What is never measured

Everything in between

Protein, secretion, contact, receptor engagement, or anything downstream of it.

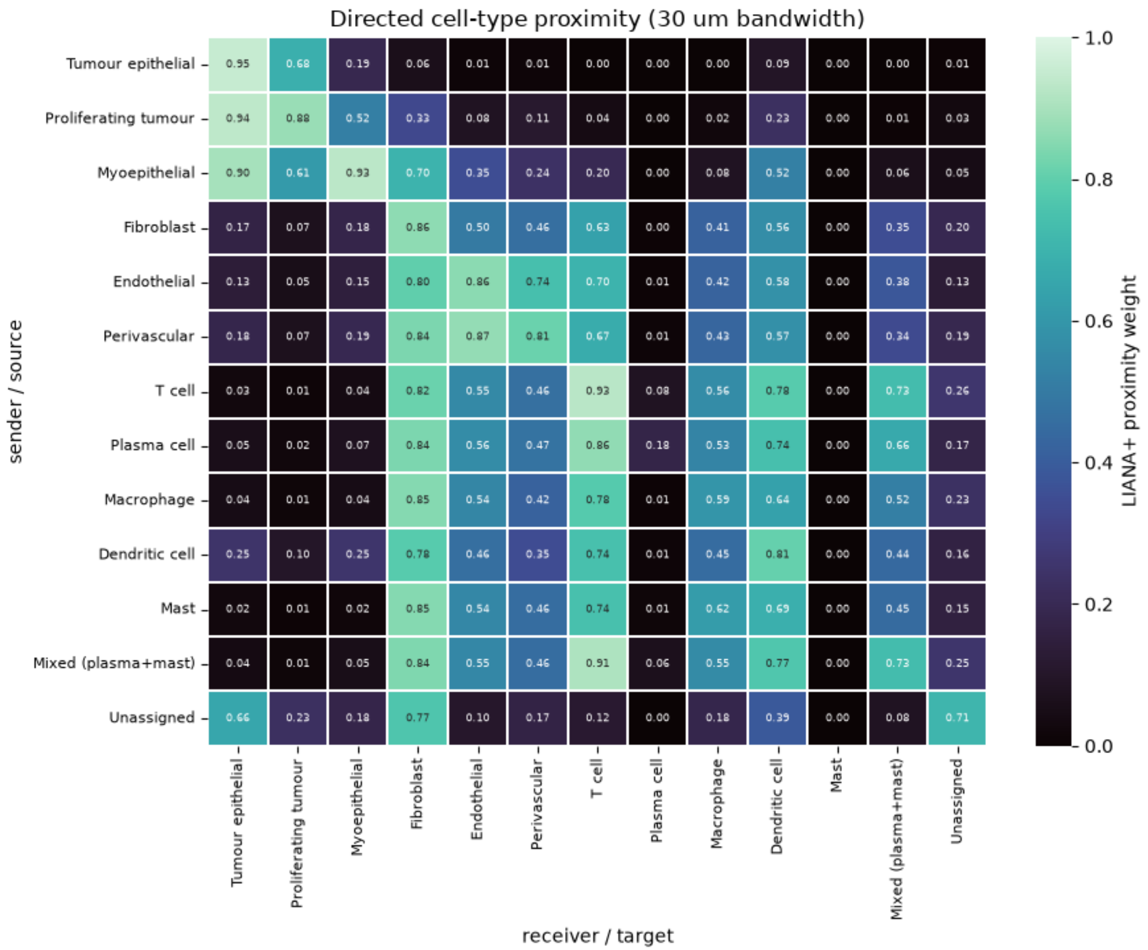
The gap is not a flaw to apologise for. It is what makes this a hypothesis generator with coordinates, rather than a result.

LIANA+ is a consensus of five scores over a database somebody curated

Score	What it rewards
CellPhoneDB	a pair expressed in a high fraction of both populations, tested against permuted labels
Connectome	a pair whose ligand and receptor are both high relative to the other populations
log2FC	a pair enriched in this sender and this receiver against everything else
NATMI	specificity: a pair carried by these two populations and few others
SingleCellSignalR	magnitude: the LRscore, a bounded product of the two expression levels

- RobustRankAggregate combines the five into magnitude_rank and specificity_rank. Smaller is stronger, and smaller is more specific.
- Complexes are encoded, so LFA-1 is ITGAL_ITGB2 and every subunit must be measured. A pair absent from the database is not refuted; it is never asked.

Space enters as a kernel over distance, and you choose the scale

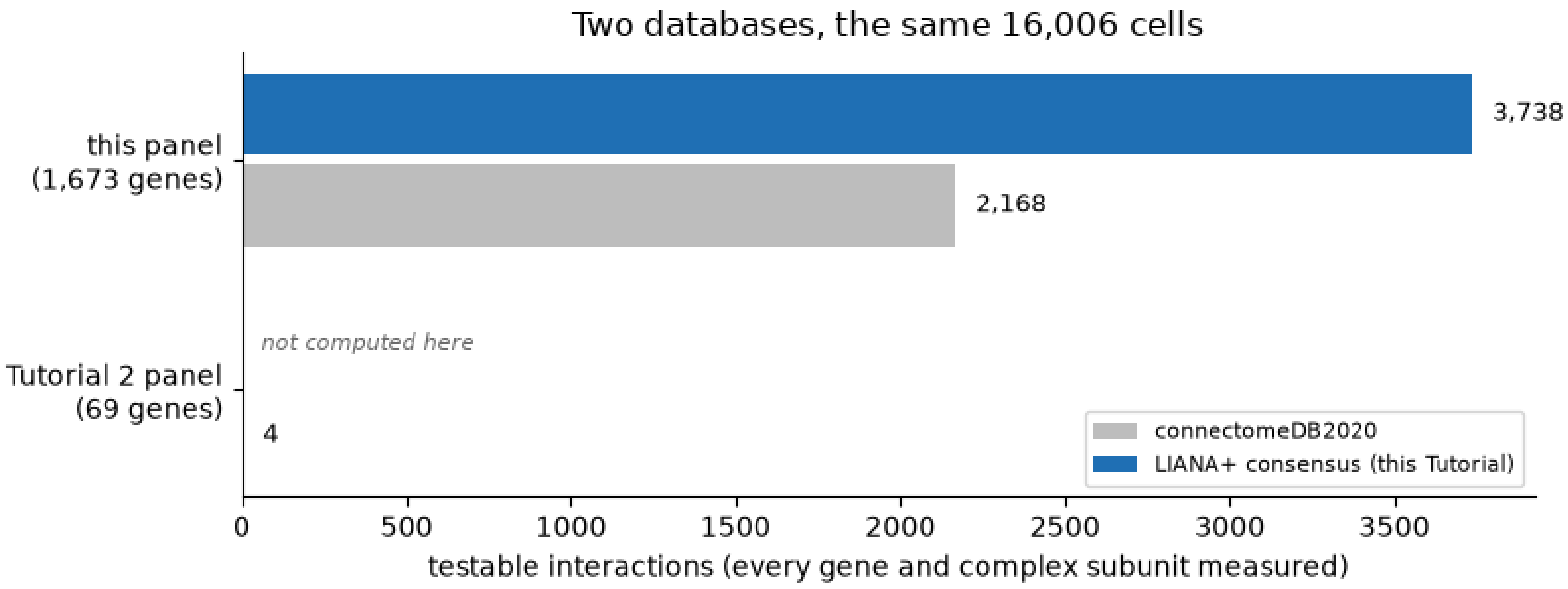


- A Gaussian kernel weights every pair by how far apart the cells are, with a cutoff below which the pair stops counting.
- Two questions, two bandwidths: 15 µm for per-cell local scores, 30 µm for between-population proximity.
- This matrix is the population-scale object: how close, on average, each sender sits to each receiver.

Tutorial 3 §4.1 · li.ut.spatial_pair_proximity, 30 µm bandwidth · directed: sender rows, receiver columns

 |

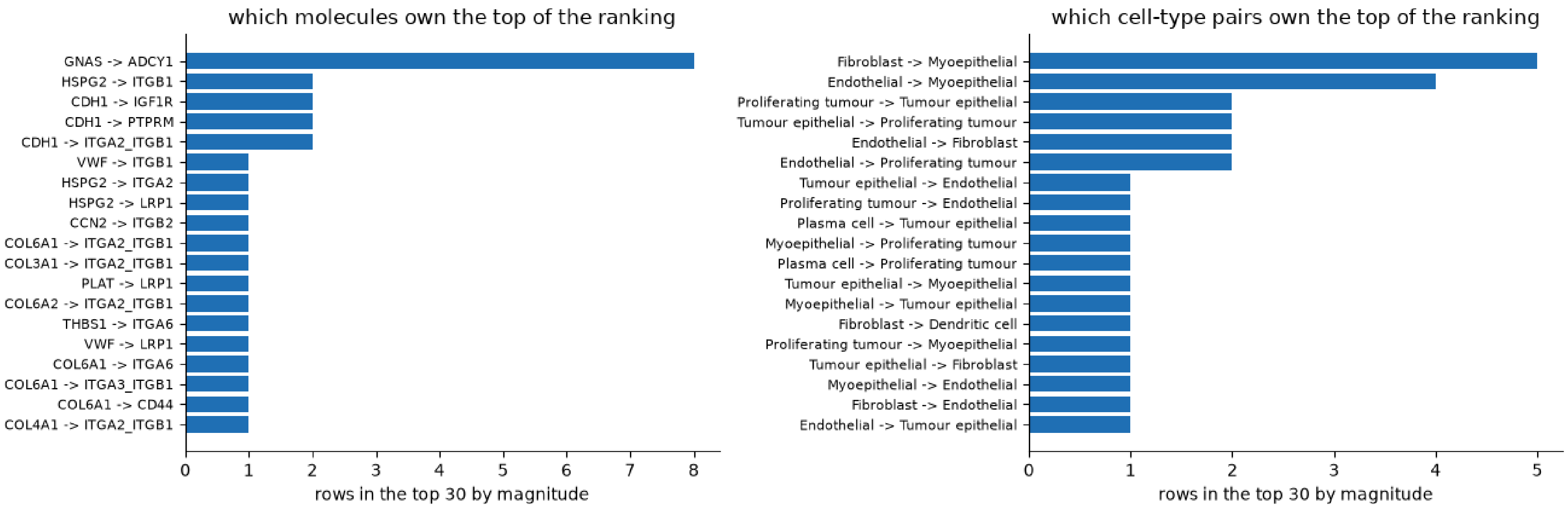
The gene panel decides which questions you are allowed to ask



- Two databases, the same 16,006 cells. The 1,673-gene panel supports 3,738 testable interactions; the 69-gene panel supports four.
- An interaction counts only if every subunit is measured. LFA-1 is ITGAL_ITGB2, not ITGAL.
- Intersect the analysis you intend to run with the probe list before you buy the panel.

Tutorial 3 §1.2 · LIANA+ consensus resource and connectomeDB2020

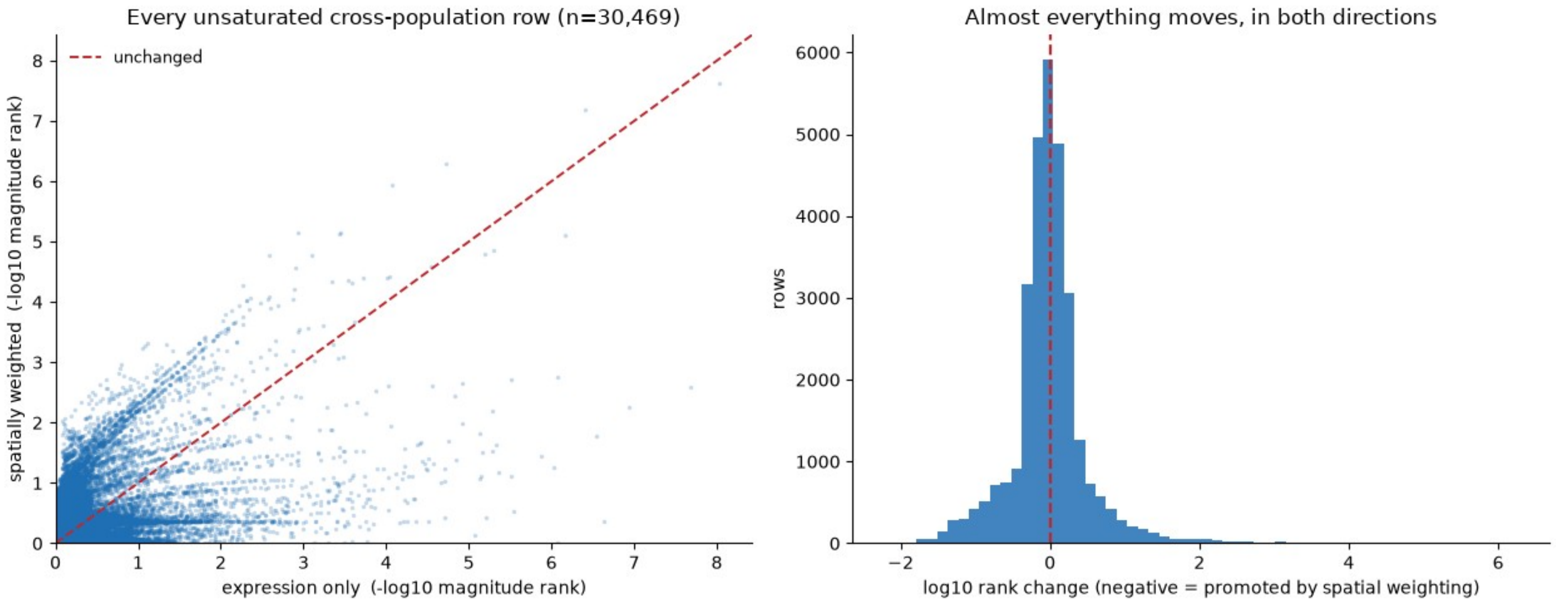
The top of a ranked ligand–receptor list follows abundance, not importance



- LIANA+ is a consensus, not a method: five scores rank-aggregated into a single ordering.
- GNAS is detected in 74% of cells, ARF1 in 57%, CDH1 in 53%. The ranking reports opportunity.
- Five statistics computed over one matrix are not five independent experiments.

Tutorial 3 §3.1 · li.mt.rank_aggregate, expr_prop 0.05, min_cells 20, 100 permutations

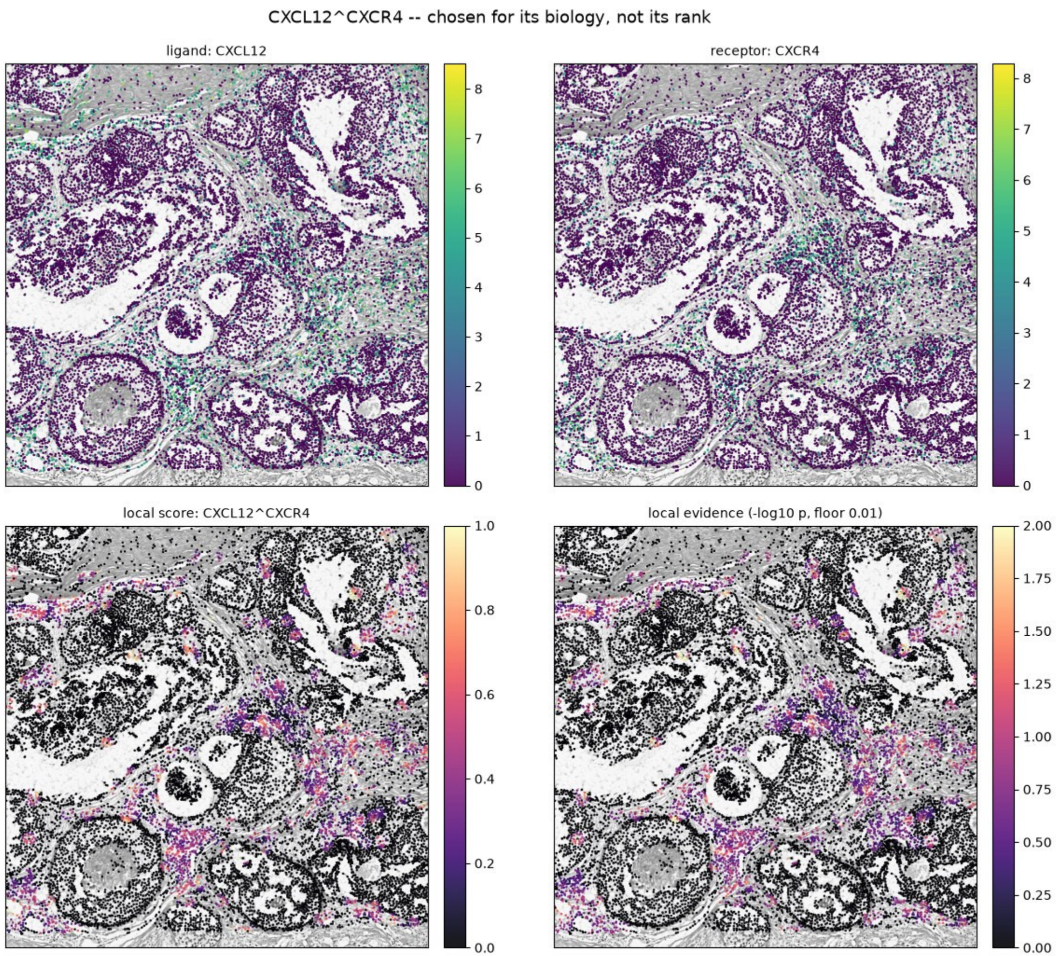
Spatial weighting re-orders almost everything. It is a constraint, not an accuracy.



- The same consensus run twice, expression only then spatially weighted: 53.7% promoted, 46.3% demoted.
- Spatial weighting is not a filter. It asks a different question, and nearly every pair moves.
- Bandwidth is not a radius: the furthest cell above cutoff sits at 2.15 × bandwidth.

Tutorial 3 §5.1 · 30,469 unsaturated cross-population rows; saturated magnitude_rank = 1.0 excluded first

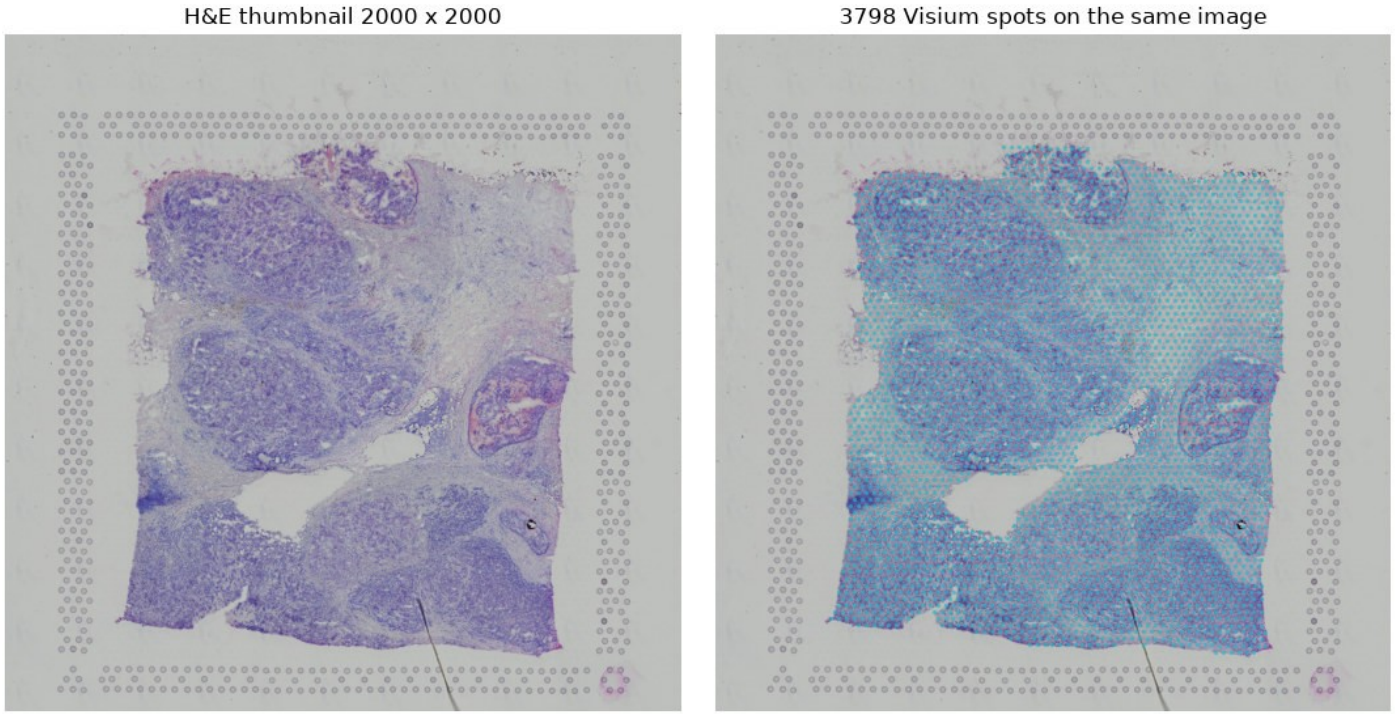
CXCL12–CXCR4 paints the same immune-exclusion geometry Tutorial 2 drew



- Per-cell local scores, not one number per population. The signal fills the stroma between the epithelial blocks and avoids the tumour nests.
- T cell ↔ dendritic cell dominates the immune half of this tumour. ICAM1–ITGAL/ITGB2 is the adhesion step of the immunological synapse.
- CD274–PDCD1 is filtered out entirely. That is a statement about detection, not about the tumour.

Tutorial 3 §6.1 · li.mt.bivariate, bandwidth 15 µm, Moran's R · LIANA+: Dimitrov et al., Nat Cell Biol 26, 1613 (2024)

Tutorial 4: can a model read biology off morphology, gene by gene?



- Every Visium spot carries two things: a patch of H&E and a vector of gene counts. The question is whether the first predicts the second.
- A pathologist already does a version of this by eye. This asks for it quantitatively, one gene at a time.
- The lineage: ST-Net and HE2RNA in 2020, then HisToGene. Production methods now start from a pathology foundation model.

Tutorial 4 §1.2 · Visium V1_Breast_Cancer_Block_A_Section_1, 3,798 spots under tissue

A Transformer is tokens, attention, and one token that reads the whole set

A token

One thing, as a vector

A word in a sentence, or here a small patch of the image.

Self-attention

Everything asks everything

Every token queries every other and mixes in whatever it finds relevant.

Position

Added, not assumed

Attention is blind to order, so where a token came from has to be told to it.

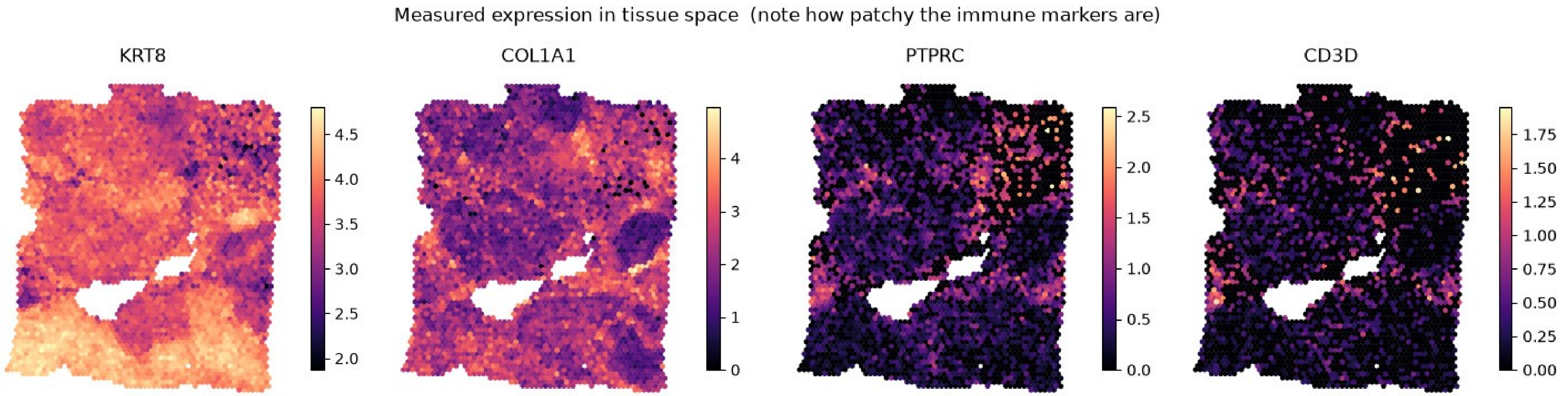
[CLS]

The one that summarises

A token belonging to no patch, which ends up carrying the answer for all of them.

- A block is attention, then a small MLP, each with a bypass around it so the signal can skip past. Stack three and that is the model.
- Nothing in that description is specific to images. Swap patches for words and it is a language model.
- You build it from nn.MultiheadAttention and nn.LayerNorm in about forty readable lines, not from a library call.

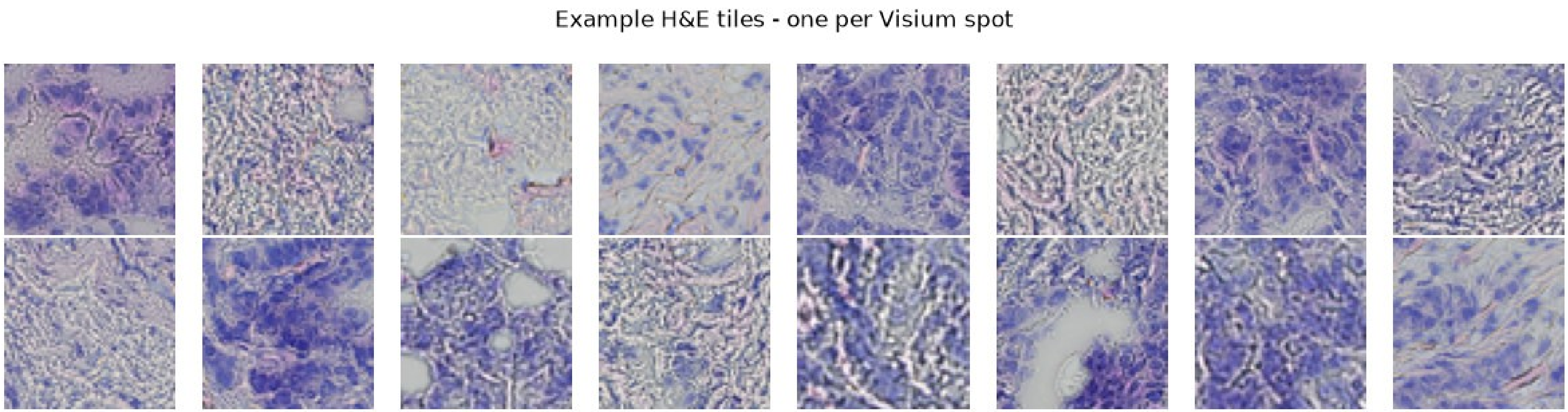
A gene you cannot detect is a gene you cannot predict



- KRT8 and COL1A1 are measured almost everywhere. PTPRC and CD3D are patchy, and mostly zero.
- Sixteen targets: twelve hand-picked breast and immune markers, topped up by dispersion, each z-scored per gene.
- The metric is one Pearson r per gene on held-out spots, not one number for the model.

Tutorial 4 §3.1 · measured expression in tissue space, log-normalised

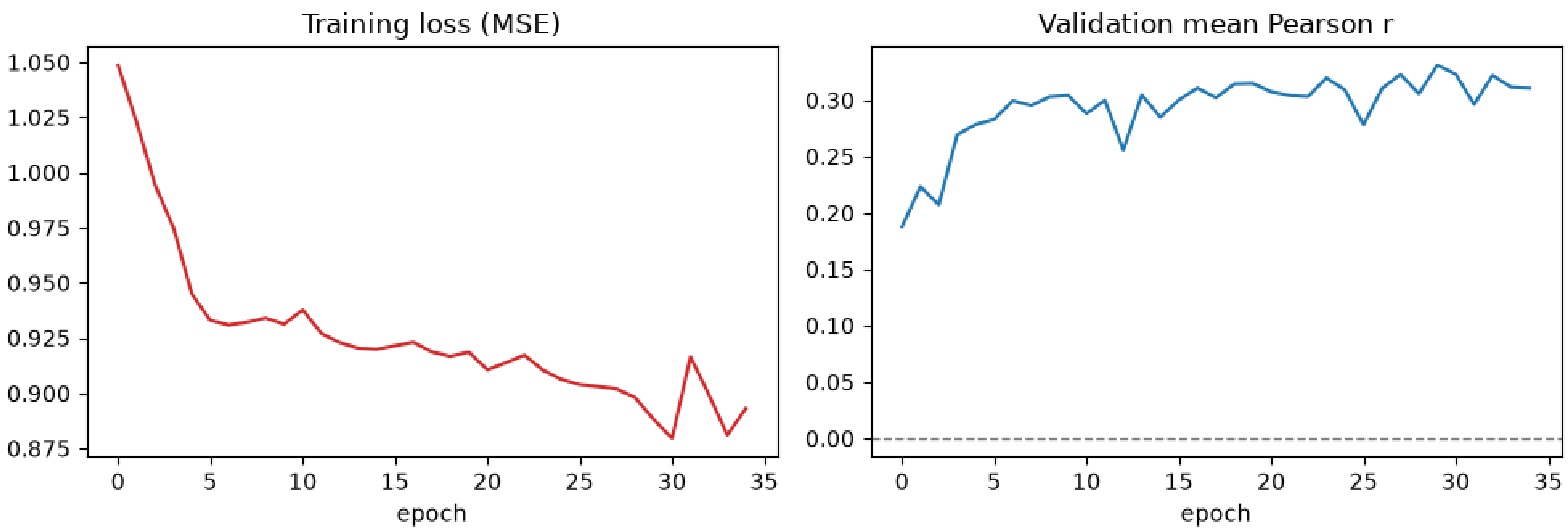
One H&E tile per spot, and full resolution is why a nucleus is 26 px and not 2



- 3,798 Visium spots, one 390 px tile each, cut from the 1.8 GB full-resolution H&E and resized to 64 px.
- On the 2,000 px thumbnail an 8 µm nucleus is 2 pixels. At full resolution it is 26.
- Cut the tiles in adata.obs_names order. Any other order silently trains the model to r ≈ 0.

Tutorial 4 §2 · Visium V1_Breast_Cancer_Block_A_Section_1, full-resolution H&E memory-mapped with tifffile

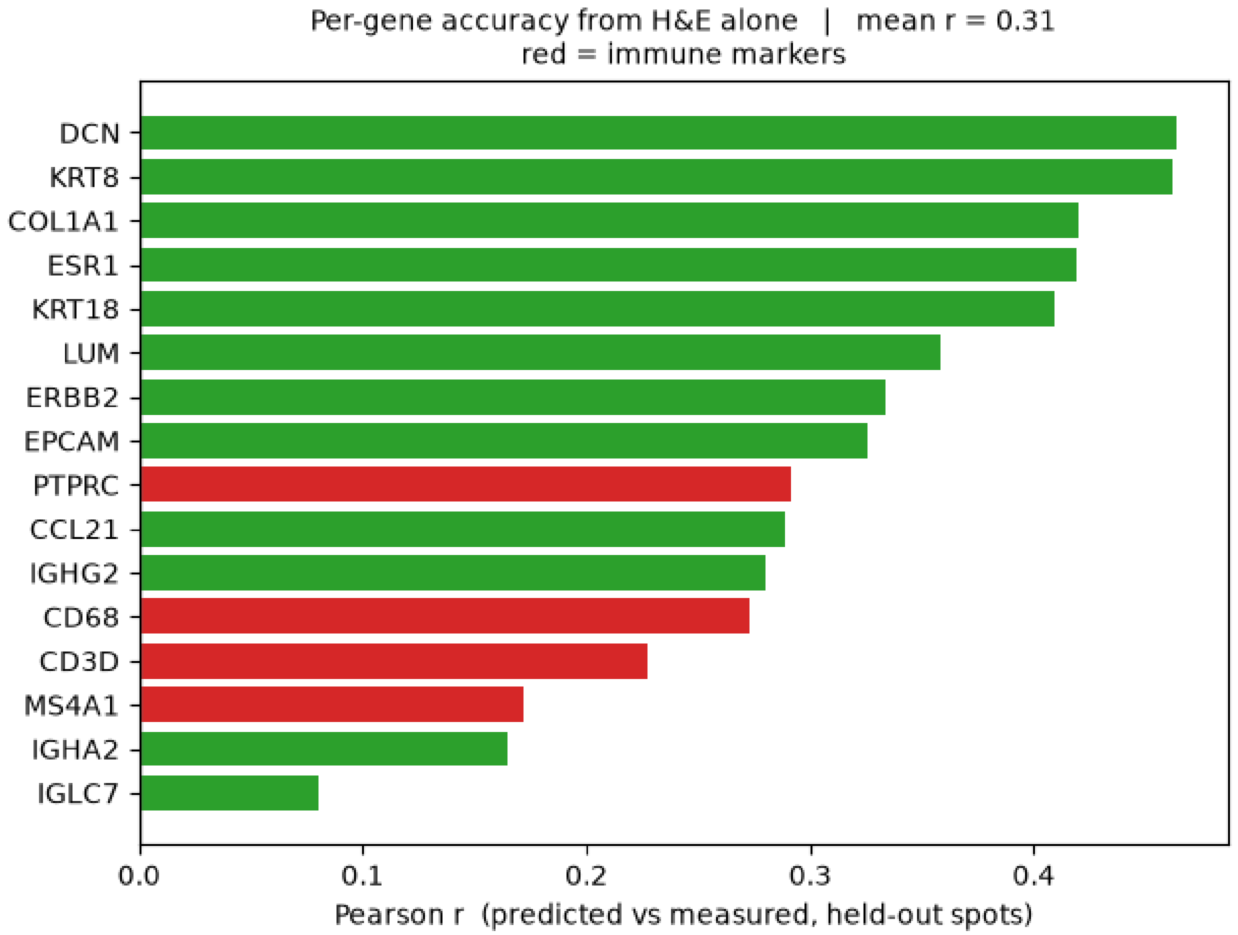
A 118,160-parameter ViT trains in three to six minutes on a free CPU runtime



- Plain PyTorch: an 8×8 patch embedding giving 64 tokens plus a [CLS], three blocks, four heads, d_model 64.
- 2,848 training and 950 held-out spots, 16 marker genes z-scored, 35 epochs.
- Spots are split at random, so a validation spot often neighbours a training spot.

Tutorial 4 §6 · derived from gml-teaching-2026, Deep_learning_04_vit_HE_spatial.ipynb

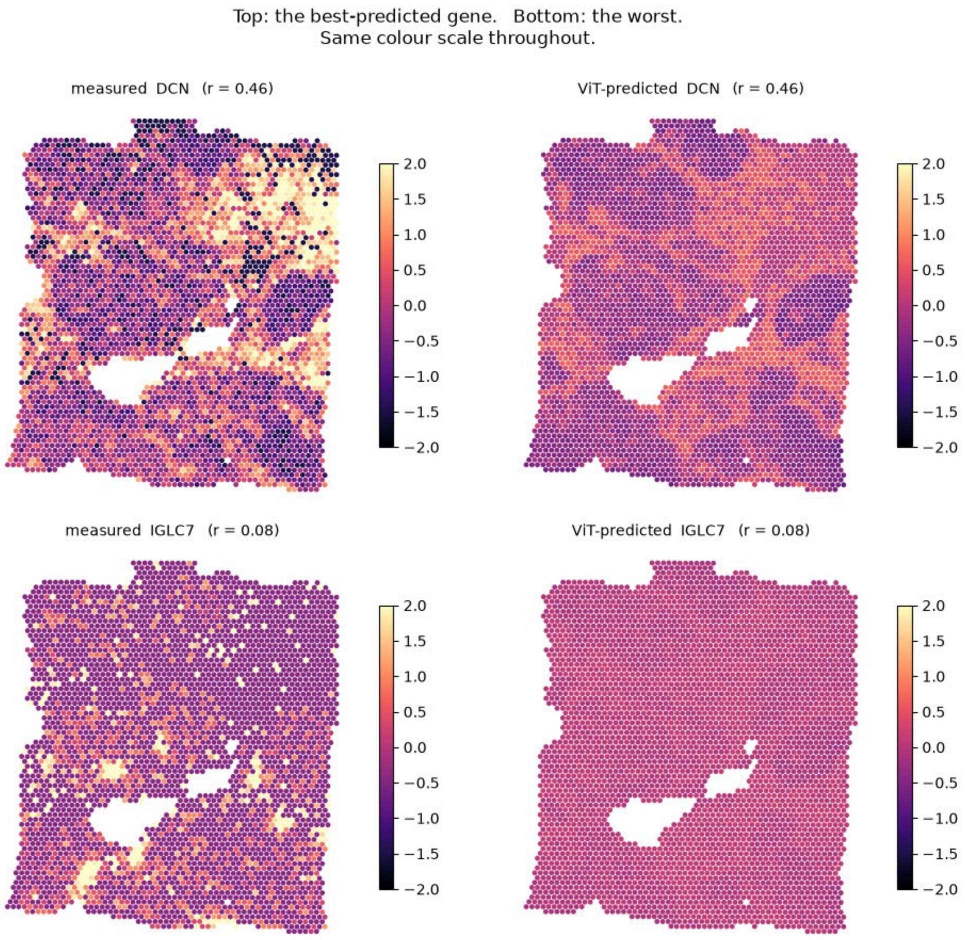
Mean r ≈ 0.31: stromal genes predict from morphology, immune markers do not



- DCN 0.47, KRT8 0.46, COL1A1 0.42. Then PTPRC 0.29, CD68 0.27, CD3D 0.23, MS4A1 0.17.
- Three reasons stacked: dropout, resolution, and morphological ambiguity.
- A small round blue cell on H&E could be a T cell, a B cell, an NK cell or a plasmacytoid dendritic cell. The information is not in the image, and no amount of training data fixes that.

Tutorial 4 §7 · Pearson r on held-out spots, one CPU run; values move slightly between runs and between CPU and GPU

It works best where you needed it least and worst where you needed it most



Tutorial 4 §7.2 · measured against ViT-predicted, held-out spots, one shared colour scale

- DCN at r = 0.46: the model reproduces the stromal architecture of the section from morphology alone.
- IGLC7 at r = 0.08: the prediction collapses to a flat band at the mean. That is what “the information is not in the image” looks like.
- Predicting expression from H&E is real and useful. Any paper claiming to read the immune microenvironment off H&E deserves this chart pointed at it.

Three methods, three failure modes, one consistent blind spot

Tutorial 2

No B-cell cluster.

About 10% of cells carry at least one B-cell gene, but graphclust does not resolve them as their own population. No label was forced.

Tutorial 3

Every immune pair ranks below the abundant adhesion biology.

CD274–PDCD1 cannot be evaluated at all: too sparse to survive the expression filter.

Tutorial 4

The four immune markers are the worst four genes.

Stromal and epithelial genes predict from morphology. PTPRC, CD68, CD3D and MS4A1 do not.

Every one of these methods has a boundary, and the boundary usually lands on the immune compartment. Knowing where it lands is the difference between using these tools and being used by them.

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Credits, licences and further reading

Data

All three datasets are generated and released by 10x Genomics. Xenium breast: Janesick, Shelansky, Gottscho et al., Nat Commun 14, 8353 (2023). The Atera whole-transcriptome preview is CC BY 4.0; Visium and Xenium are downloaded from 10x directly and not redistributed here.

Methods

LIANA+ — Dimitrov et al., Nat Cell Biol 26, 1613 (2024). squidpy, sopa, scanpy and the spatialdata family. Vision Transformer — Dosovitskiy et al., arXiv:2010.11929. Attention rollout — Abnar & Zuidema, arXiv:2005.00928.

Material

Derived from the gml-teaching-2026 course of the Genomics and Machine Learning Lab, QIMR Berghofer Medical Research Institute | The University of Queensland. Teaching material and code are MIT licensed.

github.com/xiao233333/ASI-FIMSA-workshop-2026